

Power Domination on Block-cactus Graphs*

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Abstract

The problem of monitoring an electric power system is placing as few measurement devices as possible. In graph theoretical representation, it can be considered as a variant of domination problem, namely, power domination problem. This problem is to find a minimum power domination set S of a graph $G = (V, E)$ with $S \subseteq V$ and S can dominate all vertices and edges through the observation rules according to Kirchhoff's current law and Ohm's law. In this paper, we propose a linear-time algorithm for the power domination problem on block-cactus graphs.

1 Introduction

Let $G = (V, E)$ be a finite, simple, and undirected graph where V and E are the vertex and edge sets of G , respectively. For a vertex x , let $N(x) = \{y \in V \mid (x, y) \in E\}$ and $N[x] = N(x) \cup \{x\}$. For a subset $S \subseteq V$, let $N(S) = \{v \mid v \in N(s) \text{ for some } s \in S\}$ and $N[S] = N(S) \cup S$. A subset $W \subseteq V$ of vertices is a *dominating set* if $\forall x \in V, x \in N[W]$. The DOMINATION PROBLEM on G is to find a minimum dominating set of G .

Power domination problem was first proposed to solve the problem of monitoring an electric power system by placing fewest phase measurement units (PMUs) [8]. We use graph $G = (V, E)$ to represent the electric power system with each vertex representing an electrical node. A PMU measures the vertex where it is placed and observes the incident edges and neighbors. The other observation rules are listed in the following [8].

1. Any vertex which is incident to an observed edge is observed.

2. Any edge joining two observed vertices is observed.

3. If a vertex is incident to a total of $k > 1$ edges and if $k - 1$ of these edges are observed, then all k of these edges are observed.

These observation rules observe the vertices and edges by Kirchhoff's current law and Ohm's law [1, 10]. In graph theoretical view, this problem is close to domination problem. For those vertices and edges which can be monitored, we say that they had been observed. Instead of being dominated by neighborhood relation in domination problem, the power domination depends on being observed. The POWER DOMINATION PROBLEM is to find a minimum power dominating set. The cardinality of the minimum power dominating set is called *power domination number* and is denoted as $\gamma_p(G)$ for a graph G .

The relation between power dominating set and dominating set has been proved that for any graph G , $1 \leq \gamma_p(G) \leq \gamma(G)$. Power domination problem remains NP-complete when restrict to chordal graphs [8], chordal bipartite graphs [4], and split graphs [6, 10]. It can be solved in polynomial time on some classes of graphs, *e.g.*, trees [8], interval graphs [10], block graphs [2, 14], and graphs of bounded treewidth [6, 9].

2 Block-cactus graphs

For any graph G , a vertex v is a *cut* vertex if the number of connected components is increased after removing v , and a *block* is a maximal connected subgraph without any cut vertex.

A graph is called a *block graph* if and only if its blocks are complete graphs (or cliques) and the intersection of two blocks is either empty or a cut

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vertex [7]. The definition of cactus graphs is similar. The difference is that a block in a cactus graph is a cycle. Another definition of cactus graphs is that a graph G is a *cactus graph* if every edge is part of at most one cycle [3].

In block graphs and cactus graphs, all blocks were in a restricted type (clique or cycle). We can extend them as follows. A graph G is a *block-cactus graph* if every block in G is either clique or cycle of order at least 4 [12]. The following figure shows an example.

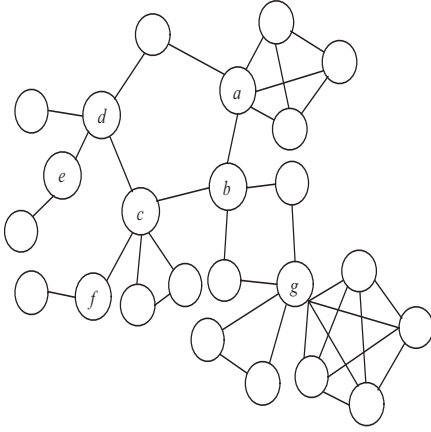


Figure 1: A block-cactus graph G .

Power domination problem on block graph had been discussed in several papers. In [1], they obtained a bound of power domination set on block graphs by dividing a block graph into block-spiders. The concept comes from solving this problem on tree with finding the number of spiders [8]. In [14], they used a different way for solving the problem. They proposed a linear color-marking-based algorithm which works on a tree-like decomposition structure of a block graph for finding a minimum power dominating set.

3 Some properties for power domination on block-cactus graphs

In this section, we present some properties and results which are helpful for solving power domination problem on block-cactus graphs.

Lemma 1 *Let G be a block-cactus graph. There exists a power dominating set of G such that all vertices in the set are cut vertices.*

Proof. Suppose S is a power domination set of G , and a vertex $v \in S$ which is not a cut vertex

in G . If v is in the block of clique, there must be a cut-vertex u with $u \in N(v)$. u is not in S , otherwise, $S - \{v\}$ would be a smaller power domination set than S . We can easily replace v by u and $S - \{v\} + \{u\}$ is still a power domination set of G , because $N[v] \subseteq N[u]$. If v is in the block of cycle, there must be a cut-vertex w which lies in the same cycle with v . w is not in S , otherwise, w may observe v by observation rule 3 and $S - \{v\}$ would be a smaller power domination set than S . We may replace v by w and $S - \{v\} + \{w\}$ is still a power domination set of G because of observation rule 3. $\square$

By Lemma 1, we only need to decide which cut vertices should be put into the power dominating set. So, we first recognize the cut vertices in a graph. The recognition of cut vertices can be done by finding a block-cutpoint graph of the input graph with the definition as follows. The *block-cutpoint graph* H of a graph G is a bipartite graph with one partite set X containing all the cut vertices and the other Y containing all the blocks such that vertex $x \in X$ is adjacent to vertex $y \in Y$ if x is a vertex in block y [13].

Figure 2 shows the block-cutpoint graph of the graph depicted in Figure 1. Note that the block-cutpoint graph of a block-cactus graph can be obtained in linear time by a depth-first search [13].

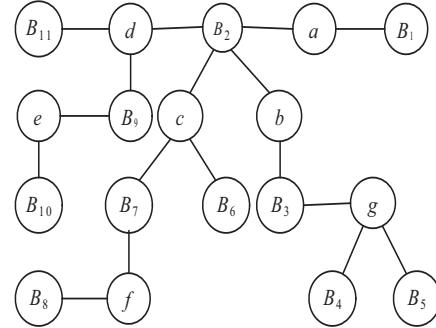


Figure 2: The block-cutpoint graph of G .

We call a block node *simple* if it connects with only one cut vertex in the block-cutpoint graph, and call a block connects with more than one cut vertex a *compound block*. Since we only consider cut vertices, we can simplify the block-cutpoint graph as follows. A simple block is still a single vertex in the modified structure. But a compound block would become a group of vertices which are the elements in the block. Instead of labeling each block as B_i for some i , we label each block by their property. For example, we use C_i to denote a block

of cycle with l vertices, and K_j to denote a block of clique with j vertices. For those concatenated K_2 and cut vertices which form a path, we compress them into a node and label it as P in the modified structure. At last, we arbitrary choose a cut vertex as the root of the tree. We call such a modified tree *BC tree* and use it as the input of our algorithm. Figure 3 shows the BC tree of the block-cutpoint graph depicted in Figure 2.

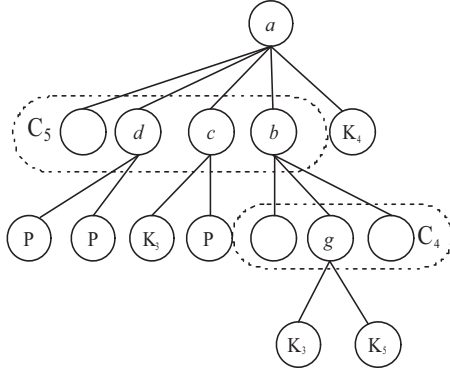


Figure 3: A BC tree of G .

In Figure 3, we can see that the vertices B_9 , e , and B_{10} in Figure 2 are compressed into a single vertex P . The simple block B_6 is relabeled as K_3 , and the compound block B_3 becomes a group of vertices of the original block in G and this group is relabeled as C_4 .

We present some properties for solving power domination problem on cliques, cycles, and paths which come from the observation rules. To ease the description, we give some notations. For those vertices in the power dominating set, we label them as *fd*, namely, *fully dominated*. A degree- k vertex with $k - 1$ incident edges being observed would be labeled as *fo*, namely, *fully observed*. Those vertices which can be observed but do not satisfy observation rule 3 would be labeled as *po*, namely, *partially observed*. It is not hard to see that a non-cut vertex must be labeled as *fo*. So that if a vertex is *fd* or *po*, it must be a cut vertex in G .

Lemma 2 *If a block which is a clique with i vertices being observed, then*

1. *at least one vertex is labeled as fd, or*
2. *$i - j - 1$, ($j > 0$), vertices are labeled as po, and j vertices are labeled as fo.*

In Case 1, if there is one vertex marked as *fd* in a clique, then it can be observed by itself and all its incident edges and neighbors are observed. Thus all vertices and edges in this block is observed. In Case 2, $i - 1$ vertices in K_i are either fully or partially observed. This means all the edges between these $i - 1$ vertices can be observed through observation rule 2. With the restriction that at least one vertex is labeled as *fo*, the fully observed vertex may let all its incident edges be observed by observation rule 3. Therefore, the remaining vertices that do not be observed can be observed through observation rule 1. All the vertices and edges in K_i can be observed in Case 2.

In the BC tree, we compress a path into a single node. Obviously, a path can be observed through rules 1 and 3. Thus we can extend Lemma 2 to the case of paths.

Lemma 3 *At least one endpoint of a path must be labeled as either fd or fo.*

Now we consider the case of cycles.

Lemma 4 *To observe a cycle C_i , the following two conditions must be satisfied.*

1. *Either at least one vertex is labeled as fd or two connected vertices are fo.*
2. *If more than one vertex is labeled as po in C_i , then C_i can be divided into several paths with each endpoint of these paths being labeled as po.*

A cycle can be observed by one vertex with *fd* or two connected vertices with *fo* through observation rule 3. Case 1 would make a bi-direction observation. So that even if there is a vertex with label *po*, the cycle can still be observed. But if there is more than one *po* vertex, the vertices in the path between two *po* vertices may not be observed because *po* vertices cannot follow the observation rules. Thus we need another vertex with label *fd* or two connected vertices with *fo* to observe the vertices and edges between the two *po* vertices.

4 An algorithm

The concept is according to Lemmas 2, 3, and 4. We set the weight of a cut vertex being 1 if it is labeled as *fd* and 0 for the other labels. We need three variables $\gamma_{pfd}(v)$, $\gamma_{pfo}(v)$, and $\gamma_{ppo}(v)$ for every cut vertex v , where $\gamma_{pfd}(v)$ is the minimum

weight power domination number for the subgraph induced by the subtree with root v and the label of v is fd ; $\gamma_{pfpo}(v)$ and $\gamma_{ppo}(v)$ are defined similar for the label of v being fo and po , respectively. Therefore, γ_p of a block-cactus graph is the minimum weight of the three variables of the root of its BC tree.

For a vertex v , let its children be u_i , $i \in \{1, \dots, s\}$. If u_j is a group of vertices which is labeled as C_k or K_k , we use x_i to represent the vertices in u_j , i from 1 to $k-1$ and we use $w(u_j)$ to denote the minimum weight of the subtrees rooted at x_i .

Lemma 5 *If u_j is a group of vertices in K_k , then $w(u_j) = \sum_{i=1}^{k-1} \min(\gamma_{pfd}(x_i), \gamma_{pfo}(x_i), \gamma_{ppo}(x_i))$. If all $\gamma_{ppo}(x_i)$ are less than $\gamma_{pfd}(x_i)$ and $\gamma_{pfo}(x_i)$, then v must be labeled as fd . Further, $\gamma_{pfo}(v) = \gamma_{ppo}(v) = \infty$.*

To make sure the power domination set is minimum, we choose the minimum one from all the subtrees. If all the minimum ones we chose are γ_{ppo} , it cannot satisfy Case 2 of Lemma 4. So v must be labeled as fd to make Case 1 be satisfied.

Lemma 6 *If u_j is a single vertex in C_k or K_k , then $w(u_j) = 0$ and v must be labeled as fd . Further, $\gamma_{pfo}(v) = \gamma_{ppo}(v) = \infty$.*

Lemma 7 *If u_j is a group of vertices in C_k , then we can compute $w(u_j)$ correctly.*

Proof. We provide the following procedure to compute $w(u_j)$.

1. For every x_i , we label it as po if $\gamma_{ppo}(x_i) < \gamma_{pfd}(x_i)$; fd if $\gamma_{ppo} = \gamma_{pfd}$ or $\gamma_{pfo}(x_i) = \gamma_{pfd}(x_i)$ when no po ; fo for the other cases.
2. We divide the path from x_1 to x_{k-1} into several paths with each endpoint of the path being labeled as po . If there is a path cannot satisfy Case 1 of Lemma 4, then we relabel one po which is near v than the other fd vertices.
3. Assume the left-most cut vertex which is labeled as po is x_a and the right-most cut vertex which is labeled as po is x_b . If both $v - x_a$ path and $v - x_b$ path cannot satisfy Case 1 of Lemma 4, v must be labeled as fd . If only one of the path may satisfy Case 1 of Lemma 4, v cannot be labeled as fo . If both two paths may satisfy Lemma 4, v cannot be labeled as po .

4. $w(u_j) = \sum \gamma(x_i)$. (According to the labels we had selected in former steps.)

The labeling determined in the above procedure makes $w(u_j)$ be the minimum one. If there is a cut vertex labeled as fd should be labeled as po to obtain a minimum weight, it must not satisfy Lemma 4. Step 1 of the procedure makes all vertices be po when $\gamma_{ppo}(x_i) < \gamma_{pfd}(x_i)$. We only relabel them into po when they cannot satisfy Case 2 of Lemma 4. Therefore, we obtain the minimum weight $w(u_j)$. $\square$

Lemma 8 *If u_j is P , then we use $c(u_j)$ to denote the child of u_j . There are three cases.*

1. *If more than one P are leaf nodes, then v must be labeled as fd . Thus, $w(u_j) = \min(\gamma_{pfd}(c(u_j)), \gamma_{pfo}(c(u_j)), \gamma_{ppo}(c(u_j)))$, and $\gamma_{pfo}(v) = \gamma_{ppo}(v) = \infty$.*
2. *If only one P is a leaf node, then $w(u_j) = \min(\gamma_{pfd}(c(u_j)), \gamma_{pfo}(c(u_j)))$. And v can only be labeled as either fd or po . Thus, $\gamma_{pfo}(v) = \infty$.*
3. *If there is no P in the leaf nodes, then $w(u_j) = \min(\gamma_{pfd}(c(u_j)), \gamma_{pfo}(c(u_j)), \gamma_{ppo}(c(u_j)))$. If more than one $\gamma_{ppo}(u_j)$ are less than $\gamma_{pfd}(u_j)$, then v can only be labeled as fd .*

Each of the cases in Lemma 8 satisfies the requirement of Lemma 3. Algorithm 1 shows the detail.

Consider Figure 1 as an example. Let the BC tree depicted in Figure 3 be the input of our algorithm. We first get $\gamma_{pfd}(g) = 1$, and $\gamma_{pfo}(g) = \gamma_{ppo}(g) = \infty$. We get the same results on vertex d and c . For vertex b , we obtain $\gamma_{pfd}(b) = 2$, and $\gamma_{pfo}(b) = 1$, $\gamma_{ppo}(b) = \infty$. Finally we obtain $\gamma_{pfd}(a) = 4$, and $\gamma_{pfo}(a) = \gamma_{ppo}(a) = \infty$. Thus $\gamma_p(G)$ is 4.

5 Conclusion

In this paper, we propose a linear-time algorithm for solving the power domination problem on block-cactus graphs. Though the time complexity of the problem on graphs of bounded treewidth is polynomial, the constant hidden in the big- O is very large. Thus finding practical algorithm for graphs of bounded treewidth like outerplanar graphs and series-parallel graphs will be interesting tasks.

Algorithm 1 Finding $\gamma_p(G)$ for block-cactus graph G

Input: The BC tree T of G with r being the root.

Output: $\gamma_p(G)$.

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1: Traverse  $T$  from leaves to  $r$ . Let  $w_v$  be the
   weight of the subtree rooted at  $v$ .
2: for every cut vertex  $v$  do
3:   for all children of  $v$ ,  $u_j$   $j \in \{1, \dots, s\}$  do
4:     if  $u_j$  is a group labeled as  $C_k$  then
5:       for all  $x_i \in \text{child vertices of } u_j$ ,  $i$  from
         1 to  $k-1$  do
6:         if  $\gamma_{ppo}(x_i) < \gamma_{pfd}(x_i)$  then
7:           label  $x_i$  as  $po$ 
8:         else if  $\gamma_{pfo}(x_i) = \gamma_{pfd}(x_i)$  then
9:           label  $x_i$  as  $fd$ 
10:        else
11:          label  $x_i$  as  $fo$ 
12:        end if
13:      end for
14:    for every pair of  $x_i$  labeled as  $po$  do
15:      if neither one  $fd$  nor two connected
         $fo$  in the path then
16:        Relabel the one which is near  $v$  as
           $fd$ 
17:      end if
18:    end for
19:     $w_{u_j} = \sum \gamma(x_i)$ , according to the label-
      ing in former steps.
20:  end if
21:  if  $u_j$  is a group labeled as  $K_k$  then
22:     $w_{u_j} = \sum_{i=1}^{k-1} \min(\gamma_{pfd}(x_i), \gamma_{pfo}(x_i), \gamma_{ppo}(x_i))$ ,
      where  $x_i \in \text{child vertices of } u_j$ .
23:    if all  $\gamma_{ppo}(x_i) < \gamma_{pfo}(x_i)$  then
24:       $v$  is labeled as  $fd$ 
25:    end if
26:  end if
27:  if  $u_j$  is a vertex  $P$  then
28:    the child vertex of  $P$  is  $c(u_j)$ .
29:     $w(u_j) = \min(\gamma_{pfd}(c(u_j)), \gamma_{pfo}(c(u_j)),$ 
       $\gamma_{ppo}(c(u_j)))$ 
30:  end if
31: end for
32:  $\gamma_{pfd}(v) = 1 + \sum w(u_j)$ 
33:  $\gamma_{pfd}(v) = \sum w(u_j)$ 
34:  $\gamma_{pfd}(v) = \sum w(u_j)$ 
35: if there is at least 1 vertex labeled as  $C_k$  or
       $K_k$  or more than 2 vertices labeled as  $po$  in
       $\{u_j\}$  or  $v$  is  $fd$  then
36:    $\gamma_{pfo}(v) = \gamma_{ppo}(v) = \infty$ 
37: end if
38: end for
39: return  $\min(\gamma_{pfd}(r), \gamma_{pfo}(r), \gamma_{ppo}(r))$ 

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